

TEAM CIPHER · 2-DAY WORKSHOP

Flutter Fundamentals

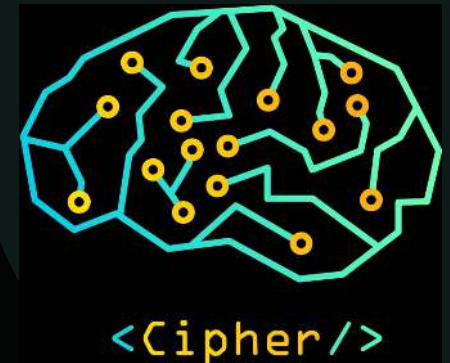
Build Your First App — from Flutter Widgets to a Working Counter App

Instructor

Alaa Elnagih

Day 1 · Flutter

Day 2 · State & UI



Two Days, One Working App

DAY 1

Flutter Basics + First Widgets

- What is Flutter, and why use it?
- Flutter vs Native — making the choice
- Flutter project structure & Widgets
- Hands-on: Profile Card app (StatelessWidget)

DAY 2

State + a Real Counter App

- StatelessWidget vs StatefulWidget
- setState — how the screen updates
- Hands-on: Simple Counter App
- One button, one increment function
- One State variable holds the count
- What's next after the workshop



Day 1

Flutter Basics & Your First Widgets

What is Flutter?

- A framework by Google to build apps for Android, iOS, web, and desktop from ONE codebase.
- Uses the Dart language — you write the UI code once, it runs everywhere.
- Flutter draws every pixel itself, so the app looks identical on every device.
- Open source, backed by Google — used by Google Pay, BMW, Alibaba, eBay Motors.

```
ElevatedButton(  
  onPressed: () {  
    print('Button tapped!');  
  },  
  child: Text('Tap here'),  
)
```

A full button in 6 lines of code.

MAKING THE CHOICE

Flutter vs Native Development

	Native	Flutter
Codebases	2 (Swift + Kotlin)	1 for everything
Cost & time	Higher	Lower
Performance	Excellent	Near-native
Newest OS features	Instant	Slight delay
App size	Smaller	Slightly larger

Best fit: startups & MVPs → Flutter · High-end 3D games → Native/Unity

Flutter Project Structure

```
my_app/
├── lib/
│   └── main.dart      ← you'll live here
├── android/
├── ios/
├── pubspec.yaml      ← name + packages
└── test/
```

- lib/ — 90% of your work happens here.
- main.dart — the entry point; contains main().
- pubspec.yaml — your project's ingredient list: name, version, and packages.

```
void main() {
  runApp(MyApp());
}
```

Everything is a Widget

- Every visible piece of the screen — text, button, image, even spacing — is a Widget.
- Widgets nest inside widgets, like Russian dolls (or Lego bricks).
- MyApp is the outer 'box' — mostly boilerplate that stays the same in every project.

Scaffold

Center

Column

CircleAvatar

Text('Alaa Elnagih')

Building It: Profile Card App

```
class ProfileCard extends StatelessWidget {
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('My Profile')),
      body: Center(
        child: Column(
          children: [
            CircleAvatar(radius: 60, ...),
            Text('Alaa Elnagih', style: TextStyle(
              fontSize: 24, fontWeight: FontWeight.bold)),
          ],
        ),
      ),
    );
  }
}
```

StatelessWidget only

- Nothing on this screen changes on its own.
- children: takes a List of widgets — order on screen follows list order.
- StatefulWidget waits for Day 2 — no reason to mix it in yet.